ADVANCED OPTIMISATION

1. a) i) The problem can be formulated as

$$\begin{aligned} \min_{P_i,W_i} \quad & \sum_{i=1}^{n} -\alpha_i W_i \log\left(1+\beta_i \frac{P_i}{W_i}\right) \\ s.t. \quad & \sum_{i=1}^{n} P_i = P_{tot}, \\ & \sum_{i=1}^{n} W_i = W_{tot}, \\ & P \succeq 0, \\ & W \succeq 0. \end{aligned}$$

Note that we have changed the maximisation in minimisation by changing the sign of the objective function.

ii) Note that

$$\begin{aligned} \text{kl_div}(\alpha_i W_i, \alpha_i(W_i+\beta_i P_i)) &= \alpha_i W_i \log\left(\frac{\alpha_i W_i}{\alpha_i(W_i+\beta_i P_i)}\right) - \alpha_i W_i + \alpha_i(W_i+\beta_i P_i) \\ &= -\alpha_i W_i \log\left(1+\beta_i \frac{P_i}{W_i}\right) + \alpha_i \beta_i P_i \\ &= -R_i + \alpha_i \beta_i P_i. \end{aligned}$$

Hence, the objective function can be rewritten as

$$\sum_{i=1}^{n} \text{kl_div}(\alpha_i W_i, \alpha_i(W_i+\beta_i P_i)) - \alpha_i \beta_i P_i.$$

b) i) We can find the distance between the two poliheadra by solving the QP

$$\begin{aligned} \min_{x_1,x_2} \quad & ||x_1 - x_2||_2^2 \\ s.t. \quad & A_1 x_1 \preccurlyeq b_1, \\ & A_2 x_2 \preccurlyeq b_2. \end{aligned}$$

Note that we have squared the objective function to make it a QP.

ii) The problem is infeasible if and only if one of the polyhedra is empty.

iii) The optimal value is zero if and only if the polyhedra intersect. Otherwise the optimal x_1 and x_2 are the points in $\mathscr{P}_1$ and $\mathscr{P}_2$, respectively, that are closest to each other.

Answers

2. a) The feasible set is the interval $[2, 4]$. The (unique) optimal point is $x^* = 2$, and the optimal value is $p^* = 5$.

b) Figure 2.1 shows the objective function (solid/blue line), the constraint function (dashed/orange line), the feasible set (shaded/grey area). The plot also indicates p^* (red/dot).

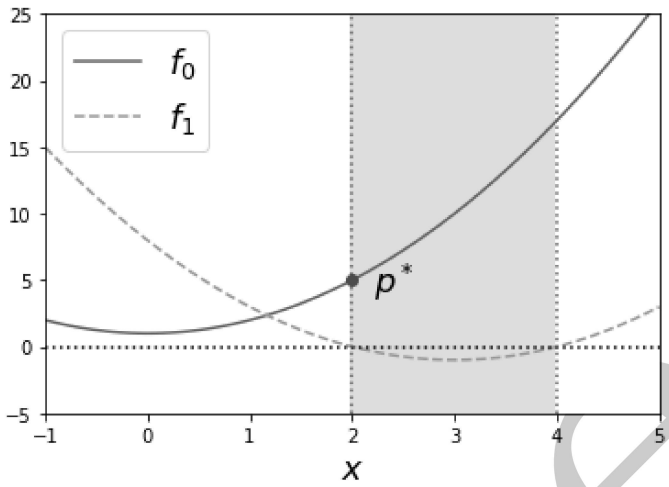


Figure 2.1 Plot for part b).

c) The Lagrangian is

$$L(x,\lambda) = (1+\lambda)x^2 - 6\lambda x + (1+8\lambda).$$

For $\lambda > -1$, the Lagrangian reaches its minimum at $\tilde{x} = \dfrac{3\lambda}{1+\lambda}$. For $\lambda \leq -1$, the Lagrangian is unbounded from below. Thus the dual function is

$$g(\lambda) = \begin{cases} \dfrac{-\lambda^2 + 9\lambda + 1}{1+\lambda} & \lambda > -1 \\ -\infty & \lambda \leq -1. \end{cases}$$

d) The Lagrange dual problem is

$$\begin{array}{ll} \max\limits_{\lambda} & \dfrac{-\lambda^2 + 9\lambda + 1}{1+\lambda} \\ s.t. & \lambda \geq 0. \end{array}$$

The dual optimum occurs at $\lambda^* = 2$, with $d^* = 5$. Figure 2.2 shows the dual function (solid/blue line) and the maximum $d^* = p^*$ (red/dot). We can directly observe that strong duality holds (as it must as Slater's constraint qualification is satisfied).

Answers

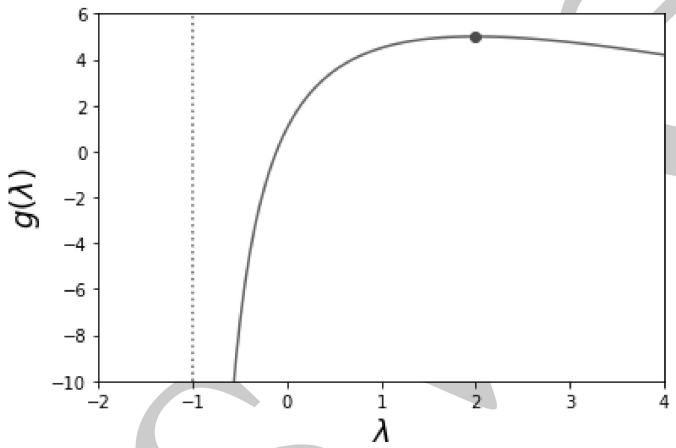


Figure 2.2 Plot for part d.ii).

Answers

3. a) i) Take two points x_1 and x_2 in the slab, i.e. $\alpha \le a^\top x_1 \le \beta$ and $\alpha \le a^\top x_2 \le \beta$. We want to show that any convex combination is inside the slab. For any $0 \le \theta \le 1$ we have

$$a^\top(\theta x_1 + (1-\theta)x_2) = \theta a^\top x_1 + (1-\theta)a^\top x_2$$

Hence,

$$\theta a^\top x_1 + (1-\theta)a^\top x_2 \le \theta\beta + (1-\theta)\beta = \beta$$

and

$$\theta a^\top x_1 + (1-\theta)a^\top x_2 \ge \theta\alpha + (1-\theta)\alpha = \alpha$$

Consequently, the point $\theta x_1 + (1-\theta)x_2$ is inside the slab.

ii) Take two points (x_1, t_1) and (x_2, t_2) in the norm cone, i.e. $||x_1|| \le t_1$ and $||x_2|| \le t_2$. We want to show that any convex combination is inside the norm cone, which means that for any $0 \le \theta \le 1$ we need to show that

$$||\theta x_1 + (1-\theta)x_2|| \le \theta t_1 + (1-\theta)t_2$$

This is easy to show

$$||\theta x_1 + (1-\theta)x_2|| \le \theta||x_1|| + (1-\theta)||x_2|| \le \theta t_1 + (1-\theta)t_2,$$

where we used the triangle inequality for the first inequality and the assumption for the second inequality.

b) i) Let $f = \max_i x_i$ and $0 \le \theta \le 1$. Then

$$\begin{aligned} f(\theta x + (1-\theta)y) &= \max_i(\theta x_i + (1-\theta)y_i) \\ &\le \theta \max_i x_i + (1-\theta)\max_i y_i = \theta f(x) + (1-\theta)f(y). \end{aligned}$$

ii) The Hessian of f is

$$\nabla^2 f(x) = \frac{1}{x_1 x_2}\begin{bmatrix} \frac{2}{x_1^2} & \frac{1}{x_1 x_2} \\ \frac{1}{x_1 x_2} & \frac{2}{x_2^2} \end{bmatrix}$$

which is positive semidefinite. Thus f is convex.

c) This can be solved by writing the Lagrangian

$$L(x, \nu) = x^\top x + \nu^\top(Ax - b)$$

and minimizing over x and ν, obtaining

$$2x + A^\top \nu = 0 \qquad Ax - b = 0$$

which gives $x = -(1/2)A^\top \nu$. Replacing this in $Ax - b = 0$ and solving for ν yields

$$\nu^* = -2(AA^\top)^{-1}b \qquad x^* = A^\top(AA^\top)^{-1}b.$$

d) The equivalent LP is

$$\begin{aligned} \min_{t,x} \quad & t \\ \text{s.t.} \quad & -t\mathbf{1} \preccurlyeq Ax - b \preccurlyeq t\mathbf{1}. \end{aligned}$$

Answers

To see the equivalence, assume x is fixed in this problem and we optimise only over t. The constraints say that $-t \leq a_k^\top x - b_k \leq t$ for all k, which implies $t \geq \max_k |a_k^\top x - b_k| = ||Ax - b||_\infty$.

Clearly, if x is fixed, the optimal value over t of the LP is $||Ax - b||_\infty$. Thus, optimising over t and x simultaneously is equivalent to the original problem.

Answers

4. a) Let $z = x^*(t)$ be the minimiser of $tf_0(x) + \phi(x)$. Thus, the minimiser satisfies $t\nabla f_0(z) + \nabla\phi(z) = 0$. Expanding we get

$$t\nabla f_0(z) + \sum_{i=1}^{m} \frac{1}{-f_i(z)} \nabla f_i(z) = 0.$$

This shows that z minimises the Lagrangian $f_0(x) + \sum_{i=1}^{m} \lambda_i(t) f_i(x)$ for

$$\lambda_i^*(t) = -\frac{1}{tf_i(z)}, \qquad i = 1, \ldots, m.$$

The associated dual function is

$$g(\lambda^*) = f_0(z) + \sum_{i=1}^{m} \lambda_i^*(t) f_i(z) = f_0(z) - \sum_{i=1}^{m} \frac{1}{t}.$$

Hence, the duality gap is

$$\frac{1}{t} \sum_{i=1}^{m} 1 = \frac{m}{t}.$$

b) The complementary condition is

$$-\lambda_i(t) f_i(z) = \frac{1}{tf_i(z)} f_i(z) = \frac{1}{t}.$$

Obviously, as $t \to \infty$, $\lambda_i(t) f_i(z) \to 0$.

c) Differentiating the centrality equation yields

$$\nabla f_0(z) + \left(t\nabla^2 f_0(z) + \nabla^2 \phi(z)\right) \frac{dx^*}{dt} = 0.$$

Thus, the tangent to the central path at $x^*(t)$ is given by

$$\frac{dx^*}{dt} = -\left(t\nabla^2 f_0(x^*(t)) + \nabla^2 \phi(x^*(t))\right)^{-1} \nabla f_0(x^*(t)).$$

Answers